PHY101:Online Quiz # 1 Physics (PHY101) Quiz Start Time: 12:30 PM, 27 May 2021

Question # 10 of 10 (Start time: 12:38:03 PM, 27 May 2021) Total Marks: 1

If force of friction is negligible, then acceleration of two freely falling objects of different masses is:

Select the correct option

- ○ Variable
- ○ None of these
- ● Smaller acceleration for smaller mass
- ○ The same

Question # 9 of 10 (Start time: 12:37:01 PM, 27 May 2021) Total Marks:

In simple harmonic motion, the magnitude of the acceleration is

Select the correct option

○	greatest when the velocity is greatest
●	proportional to the displacement
○	inversely proportional to the displacement
○	constant

PHY101:Online Quiz # 1 Physics (PHY101) Quiz Start Time: 12:30 PM, 27 May 2021

Question # 9 of 10 (Start time: 12:37:01 PM, 27 May 2021) Total Marks: 1

In simple harmonic motion, the magnitude of the acceleration is

Select the correct option

- ○ greatest when the velocity is greatest
- ● proportional to the displacement
- ○ inversely proportional to the displacement
- ○ constant

PHY101:Online Quiz # 1 Physics (PHY101) Quiz Start Time: 12:30 PM, 27 May 2021

Question # 8 of 10 (Start time: 12:36:17 PM, 27 May 2021) Total Marks: 1

A particle, held by a string whose other end is attached to a fixed point C, moves in a circle on a horizontal frictionless surface. If the string is cut, the angular momentum of the particle about the point C:

Select the correct option

- [x] Does not change
- [ ] Changes direction but not magnitude
- [ ] Decreases
- [ ] Increases

PHY101:Online Quiz # 1 Physics (PHY101) Quiz Start Time: 12:30 PM, 27 May 2021

Question # 7 of 10 (Start time: 12:35:08 PM, 27 May 2021) Total Marks: 1

A body will be in translational equilibrium if:

Select the correct option

- ( ) $\Sigma P = 0$
- (•) $\Sigma F = 0$
- ( ) $\Sigma W = 0$
- ( ) $\Sigma T = 0$

PHY101:Online Quiz # 1 Physics (PHY101)

Quiz Start Time: 12:30 PM, 27 May 2021

Question # 6 of 10 (Start time: 12:34:36 PM, 27 May 2021)

Total Marks: 1

A stone is thrown up from the surface of earth, the it reaches a maximum height, kinetic energy is equal to:

Select the correct option

○	½ mv
○	mgh
○	2mgh
●	Zero

PHY101 [illegible] Quiz # 1 Physics (PHY101) Quiz Start Time: 12:30 PM, 27 May 2021

Question # 4 of 10 (Start time: 12:32:37 PM, 27 May 2021) Total Marks: 1

If a wheel turns with constant angular speed then:

Select the correct option

- ○ each point on its rim moves with constant velocity
- ○ each point on its rim moves with constant acceleration
- ○ the wheel turns through equal angles in equal times
- ● the angle through which the wheel turns in each second increases as time goes on

PHY101:Online Quiz # 1 Physics (PHY101) Quiz Start Time: 12:30 PM, 27 May 2021

Question # 3 of 10 (Start time: 12:31:56 PM, 27 May 2021) Total Marks: 1

Whenever an object strikes a stationary object of equal mass:

Select the correct option

- ○ the first object must stop
- ○ the two objects cannot stick together
- ● none of these
- ○ the collision must be elastic

PHY101:Online Quiz # 1 Physics (PHY101)

Quiz Start Time: 12:30 PM, 27 May 2021

Question # 2 of 10 (Start time: 12:31:17 PM, 27 May 2021)

Total Marks: 1

An object attached to one end of a spring makes 20 complete oscillations in 10 s. Its period is:

Select the correct option

- ○ 0.5 s
- ○ 0.5Hz
- ● 2 s
- ○ 2Hz

Question # 1 of 10 (Start time: 12:30:41 PM, 27 May 2021)

Total Marks: 1

The center of mass of Earth's atmosphere is:

Select the correct option

- ○ near the outer boundary of the atmosphere
- ● near the center of Earth
- ○ a little less than halfways between Earth's surface and the outer boundary of the atmosphere
- ○ near the surface of Earth

Question # 10 of 10 (Start time: 12:17:45 PM, 27 May 2021) Total Marks: 1

If the distance between all pairs of particles of the body do not change by applying a force then the body is said to be

Select the correct option

- rigid
- small
- large
- flexible

PHY101:Online Quiz # 1 Physics (PHY101)

Quiz Start Time: 12:08 PM, 27 May 2021

Question # 8 of 10 (Start time: 12:15:38 PM, 27 May 2021)

Total Marks: 1

Two skaters toss a basketball back and forth on frictionless ice. Which of the following does not change?

Select the correct option

- the momentum of the system consisting of both skaters and the basketball
- the momentum of the system consisting of one skater and the basketball;
- the momentum of the basketball;
- the momentum of an individual skater;

Question # 9 of 10 (Start time: 12:16:49 PM, 27 May 2021) Total Marks: 1

A/an ———— is the basic reason to change in the motion of an object according to Newton's second law of motion.

Select the correct option

○	acceleration
○	change in velocity
●	net force
○	decrease in inertia

Question # 7 of 10 (Start time: 12:14:28 PM, 27 May 2021) Total Marks: 1

Swimming becomes possible because of:

Select the correct option

○	First law of motion
○	Second law of motion
●	Third law of motion
○	Law of torque

Question # 7 of 10 (Start time: 12:14:28 PM, 27 May 2021) Total Marks: 1

Swimming becomes possible because of:

Select the correct option

- ○ First law of motion
- ○ Second law of motion
- ● Third law of motion
- ○ Law of torque

PHY101:Online Quiz # 1 Physics (PHY101) Quiz Start Time: 12:08 PM, 27 May 2021

Question # 6 of 10 (Start time: 12:13:41 PM, 27 May 2021) Total Marks: 1

A sphere of moment on inertia I rolls down an inclined plane without slipping. The ratio of the rotational kinetic energy to the transitional kinetic energy is nearly:

Select the correct option

●	5/7
○	7/5
○	2/5
○	5/2

PHY101:Online Quiz # 1 Physics (PHY101)

Quiz Start Time: 12:08 PM, 27 May 2021

Question # 5 of 10 (Start time: 12:12:38 PM, 27 May 2021)

Total Marks: 1

The dimensional units of ratio of work and power is:

Select the correct option

- L
- F
- J
- T

PHY101:[illegible] # 1 Physics (PHY101) Quiz Start Time: 12:08 PM, 27 May 2021

Question # 5 of 10 (Start time: 12:12:38 PM, 27 May 2021) Total Marks: 1

The dimensional units of ratio of work and power is:

Select the correct option

○	L
○	F
○	J
●	T

PHY101:Online Quiz # 1 Physics (PHY101)

Quiz Start Time: 12:08 PM, 27 May 2021

Question # 4 of 10 (Start time: 12:11:05 PM, 27 May 2021)

Total Marks: 1

Add two vectors of length 4 m & 5 m but their orientation is not known. The length after addition of these two vectors will be:

Select the correct option

○	between 9 m and 1 m
○	9 m
○	between 9 m and 5
○	Less than 1 m

Correct

Question # 3 of 10 (Start time: 12:09:46 PM, 27 May 2021) Total Marks: 1

If force of friction is negligible, then acceleration of two freely falling objects of different masses is:

Select the correct option

- None of these
- Smaller acceleration for smaller mass
- Variable
- The same

uestion # 2 of 10 (Start time: 12:00:00 PM, 27 May 2021) Total Marks: 1

the projectile path is known as its

Select the correct option

- range
- trajectory
- curve
- time of action

PHY101:Online Quiz # 1 Physics (PHY101)

Quiz Start Time: 12:08 PM, 27 May 2021

Question # 1 of 10 (Start time: 12:08:01 PM, 27 May 2021)

Total Marks: 1

Q. No. 1: The application/s of dimensional analysis is/are:
i. To convert a physical quantity from one system of units to another.
ii. To check the dimensional correctness of a given equation.
iii. Establish a relationship between different physical quantities in an equation.

Select the correct option

○	i only
○	ii & iii only
●	i, ii & iii
○	i & iii only